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Bo Li, Peiwen Tan, Jie Jia & Xinyan Chen

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Multi-Scale and Edge-Aware IFGNet for Precise Skin Lesion Segmentation in High-Resolution Dermoscopic Images

Bo Li[1], Peiwen Tan[1,*], Jie Jia[2], and Xinyan Chen[3]

[1]the College of Electronic Information, Shanghai Dianji University, Shanghai, 201306, China
[2]the College of Electronic Information, Shanghai Dianji University, Shanghai, 201306, China
[3]the College of Electronic Information, Shanghai Dianji University, Shanghai, 201306, China
[*]23600003100214@st.sdju.edu.cn

ABSTRACT

Accurate segmentation of dermoscopic images is essential for early melanoma diagnosis, yet current methods remain limited. CNN-based models capture local details but lack global context, whereas Transformer-based approaches model long-range dependencies but often lose fine structures; both struggle with blurred boundaries and scale variations. To address these challenges, we propose IFGNet, a hybrid segmentation framework that integrates CNN–Transformer synergy, multi-scale convolution, and boundary-aware decoding. Specifically, our design combines local–global feature fusion, large-kernel parallel convolutions without dilation, and a boundary refinement strategy to enhance lesion consistency. Extensive experiments on the ISIC 2016, ISIC 2017, and ISIC 2018 benchmarks demonstrate that IFGNet consistently surpasses state-of-the-art methods in Dice and IoU. These results highlight the effectiveness of IFGNet for accurate high-resolution skin lesion segmentation, and suggest its potential for clinical computer-aided melanoma diagnosis.

Keywords: Skin Lesion Segmentation, Attention Mechanism, Multi-Scale Feature Extraction

Introduction

Skin lesion segmentation in dermoscopic images is essential for computer-aided diagnosis, particularly for early melanoma detection, as accurate segmentation directly influences clinical outcomes and treatment planning[1,2]. Dermoscopy provides a non-invasive, high-resolution view of skin lesions, making precise segmentation critical for diagnosis, therapy selection, and disease monitoring

In recent years, deep learning, especially convolutional neural networks (CNNs), has led advances in medical image segmentation by efficiently capturing local textures and boundary details[3]. However, CNNs have limited receptive fields, restricting global context modeling and resulting in suboptimal segmentation for large or irregular lesions[4]. Transformer-based architectures effectively capture long-range dependencies and global semantics[5,6],but often fail to retain fine-grained boundary structures essential for precise lesion delineation[7]. Furthermore, dermoscopic images present additional challenges including low-contrast boundaries, irregular morphologies, and high inter-patient variability[8].

Hybrid CNN–Transformer models have been explored to integrate local detail preservation with global semantic reasoning[9]. However, these approaches typically rely on dilated convolutions for multi-scale aggregation, introducing grid artifacts and reducing boundary accuracy[10–12]. Current decoding strategies also lack explicit boundary guidance, limiting sharp contour reconstruction[13]. Recent multi-scale fusion methods, such as MLCA2F for COVID-19 lesion segmentation[14] and BG-3DM2F for Alzheimer's diagnosis[15], demonstrate the effectiveness of attention-based feature fusion. Similarly, multilayer fusion networks have been applied to skin lesion classification[16] and breast abnormality detection[17]. However, these methods primarily focus on feature fusion strategies without explicit mechanisms for low-contrast boundary enhancement. Notably, Umirzakova *et al.* showed that explicit gradient guidance significantly improves low-contrast lesion segmentation in MRI[18], suggesting that similar strategies could benefit dermoscopic image analysis.

To address these limitations, we propose IFGNet, a hybrid segmentation network integrating three key innovations:

- Dual-Path Strategy Block (DPSB): fuses local and global features using coordinate–Inception attention (CAI) to capture anisotropic directional characteristics, addressing the limitation of conventional CNNs in global context modeling
- Parallel Multi-kernel Convolution (PMC): extracts multi-scale contextual information using depthwise separable convolutions with multiple kernel sizes, avoiding grid artifacts associated with dilated convolutions while maintaining

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computational efficiency.

- Decoding Enhancement Strategy (DES): incorporates explicit gradient injection, boundary attention, and semantic consistency constraints to enhance delineation of low-contrast edges and improve multi-resolution coherence.

Distinct from existing multi-scale fusion approaches, IFGNet emphasizes explicit boundary modeling through gradient-guided refinement rather than implicit attention mechanisms, and employs lightweight anisotropic pooling to capture directional features characteristic of skin lesion asymmetry. These design choices enable robust segmentation across diverse lesion sizes, shapes, and boundary clarity levels.

Extensive experiments on ISIC 2016, ISIC 2017, and ISIC 2018 benchmarks demonstrate that IFGNet consistently outperforms state-of-the-art methods in Dice coefficient and IoU.These results highlight the effectiveness of IFGNet for accurate high-resolution skin lesion segmentation and suggest its potential for integration into clinical computer-aided melanoma diagnosis workflows.

Related Works

CNN-based Methods

CNNs have long been the backbone of medical image segmentation due to their strong capability in capturing local structures, textures, and fine edges. Early fully convolutional networks (FCNs)[19] enabled end-to-end pixel-wise prediction, while U-Net[20] introduced skip connections to combine high-resolution spatial features with deep semantics, forming the foundation for subsequent variants. U-Net++[3], UNet 3+[21], and Attention U-Net[22] further improve multi-scale feature fusion and cross-level integration, though they still do not fully resolve local-global coordination. Multi-scale methods like FPN[23] and PSPNet[11] aggregate hierarchical features but often rely on dilated convolutions, leading to grid artifacts and reduced boundary precision. Recent works incorporate attention mechanisms such as SE[24], ECA[25], and CBAM[26] to adaptively recalibrate feature responses and enhance edge preservation.

Despite these advances, CNN-based methods face fundamental limitations that motivate our work. First, their limited receptive fields restrict global context modeling, resulting in suboptimal segmentation for large or irregular lesions. Second, standard CNN architectures rely heavily on dilated convolutions for multi-scale aggregation, which introduce gridding artifacts when dilation rates are large: the kernel samples sparse, discrete points rather than continuous regions, causing local feature discontinuities and jagged boundaries[10]. Third, standard pooling operations in CNNs treat spatial dimensions isotropically, aggregating features uniformly across all directions. However, skin lesions frequently exhibitasymmetric morphology and directional texture patterns (e.g., pigment networks, streaks) that are better captured by anisotropic processing. The inability to model these directional characteristics limits boundary accuracy for irregular lesions.

Transformer-based Methods

Transformers have been adopted to overcome the limited receptive field of CNNs by modeling long-range dependencies. TransUNet[4] integrates a Vision Transformer into a U-Net-like encoder-decoder, improving segmentation in complex backgrounds. UNETR[6] extends Transformer encoders to 3D volumetric images, capturing inter-slice dependencies. Swin-Unet[27] introduces hierarchical shifted windows for efficient computation and high-resolution support. Hybrid CNN–Transformer models such as TransFuse[28] combine CNN features with Transformer tokens to preserve local texture while modeling global semantics.

However, Transformer-based and hybrid approaches exhibit several critical limitations. First, Transformers often fail to retain fine-grained boundary details due to patch tokenization and downsampling, which is essential for precise lesion delineation. Second, many hybrid methods loosely couple local and global feature extraction through simple concatenation or attention-based aggregation, lacking explicit mechanisms for enhancing low-contrast boundaries. Consequently, subtle lesion borders with intensity gradients similar to surrounding skin are often missed. Third, Transformer components incur significant computational overhead due to self-attention mechanisms with quadratic complexity relative to sequence length, limiting real-time clinical deployment for high-resolution dermoscopic images. Fourth, current decoding strategies in hybrid models often overlook explicit boundary guidance and cross-scale semantic consistency, limiting sharp contour reconstruction.

These complementary limitations of CNNs (restricted receptive fields, gridding artifacts, isotropic processing) and Transformers (boundary detail loss, implicit boundary modeling, computational inefficiency) motivate our unified approach in IFGNet.

Multi-scale Feature Modeling

Multi-scale feature extraction is crucial for lesions of varying sizes and complex morphologies. PSPNet[11] and DeepLabv3+[12] aggregate contextual information at multiple scales using dilated convolutions or pyramid pooling, but these approaches introduce gridding artifacts that degrade boundary smoothness. FPN[23] fuses hierarchical features through top-down pathways,

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while Inception modules[29] employ parallel convolutions with varying kernel sizes to capture diverse receptive fields. Recent advances by Bakkouri and Afdel demonstrate sophisticated attention-based fusion: MLCA2F[14] uses multi-level cross-attention for CT lesion segmentation, BG-3DM2F[15] applies bidirectional gating for 3D multimodal fusion, and multilayer feature fusion networks[16] achieve effective skin lesion classification. However, these methods primarily rely on isotropic pooling and implicit boundary learning, lacking explicit mechanisms for directional feature capture and low-contrast edge enhancement that are critical for dermoscopic images. IFGNet addresses these gaps through anisotropic cross-shaped pooling and explicit gradient-guided refinement, detailed in Section Methods.

Boundary-aware and Decoder Enhancement

and decoder-side enhancements are critical for high-resolution segmentation.Boundary Attention[30] generates boundary probability maps through dedicated edge-detection branches, while Gradient Guidance[31] incorporates Sobel or Laplacian features to emphasize edge structures. Semantic consistency loss[32] enforces multi-resolution prediction coherence through deep supervision. Recent evidence from MRI lesion segmentation[18] demonstrates that explicit gradient injection significantly improves low-contrast boundary detection, yet this strategy remains underexplored in dermoscopic image analysis. Existing approaches typically apply these mechanisms independently, limiting their effectiveness on subtle lesion borders. IFGNet unifies edge-aware augmentation, gradient injection, and semantic consistency within a single decoding module, enabling robust boundary refinement for challenging dermoscopic images with noisy or low-contrast lesion boundaries.

Methods

Dermoscopic image segmentation presents unique challenges that distinguish it from general medical image analysis: (1) lesions exhibit extreme variability in size (mm to cm scale), shape (regular to highly irregular), and boundary characteristics (sharp to diffuse); (2) subtle intensity gradients at lesion periphery often fall below human visual detection thresholds; (3) artifacts (hair, gel) frequently occlude critical diagnostic regions; and (4) high-resolution imaging (10–25 μm/pixel) demands computationally efficient processing. These requirements motivate our unified architecture that explicitly addresses multi-scale representation, global-local coordination, and boundary refinement—capabilities that existing methods achieve only partially or at prohibitive computational cost. Multi-scale methods (PSPNet, DeepLabV3+) employ dilated convolutions or pyramid pooling for contextual aggregation but introduce gridding artifacts and lack explicit boundary guidance. Edge-aware networks (PraNet, Boundary Attention) focus on boundary detection but treat multi-scale feature extraction as secondary, typically through simple skip connections. Hybrid CNN-Transformer architectures (TransUNet, Swin-UNet) achieve global receptive fields through self-attention but incur quadratic complexity $\mathcal{O}(n^2)$, lose fine-grained details through patch tokenization, and lack mechanisms for low-contrast boundary enhancement.Attention-augmented CNNs (Attention U-Net, SE-Net) recalibrate channel responses but apply isotropic pooling that fails to capture directional asymmetry characteristic of malignant lesions.

IFGNet is primarily composed of three key modules: the dual-path strategy module, the parallel multi-core convolution module, and the decoding enhancement strategy module. The dual-path strategy module captures fine-grained local textures through residual paths while modeling global semantic information via contextual paths, thereby balancing lesion details and overall contour. The parallel multi-core convolution module extracts features at different scales through multi-branch convolution operations, enhancing the network's adaptability to lesions of varying sizes and mitigating the grid effect caused by traditional hollow convolution. The decoding enhancement strategy module improves boundary precision and local detail restoration during the decoding phase through boundary enhancement, gradient steering, and spatial attention mechanisms. These modules work synergistically to achieve efficient feature extraction and fine segmentation of complex dermatoscopic images. These mechanisms collectively address the accuracy-efficiency trade-off that constrains existing methods.The overall architecture of IFGNet is illustrated in Figure 1. The network follows a conventional encoder–decoder paradigm with skip connections to facilitate effective multi-scale feature propagation.

Overview of the IFGNet architecture

The main architecture of IFGNet is shown in Figure 1(a).The encoder achieves multi-level feature representation through progressive downsampling. Each layer incorporates an enhanced encoder module (Figure 1(b)), where DPSB and PMC embeddings are processed via convolution operations to enable simultaneous modeling of local texture and global semantics. The standard encoder module (Figure 1(c)) primarily utilizes dual convolution blocks combined with PMC, ensuring multi-scale information is adequately expressed in shallow layers while providing rich contextual support for higher-level semantic processing. The decoder path (Figure 1(d)) progressively restores spatial resolution through up-sampling, while the DES module enhances boundary awareness by injecting boundary and gradient cues to achieve fine reconstruction. The ConvBlock module (Figure 1(e)), composed of 3×3 convolution, BN, and ReLU, serves as the fundamental building block for both encoder and decoder, ensuring training stability and feature representation. Through long-range skip connections, fine features from the

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encoder's lower layers are directly transmitted to the decoder, balancing local details and global semantics to promote feature reuse and mitigate gradient vanishing.

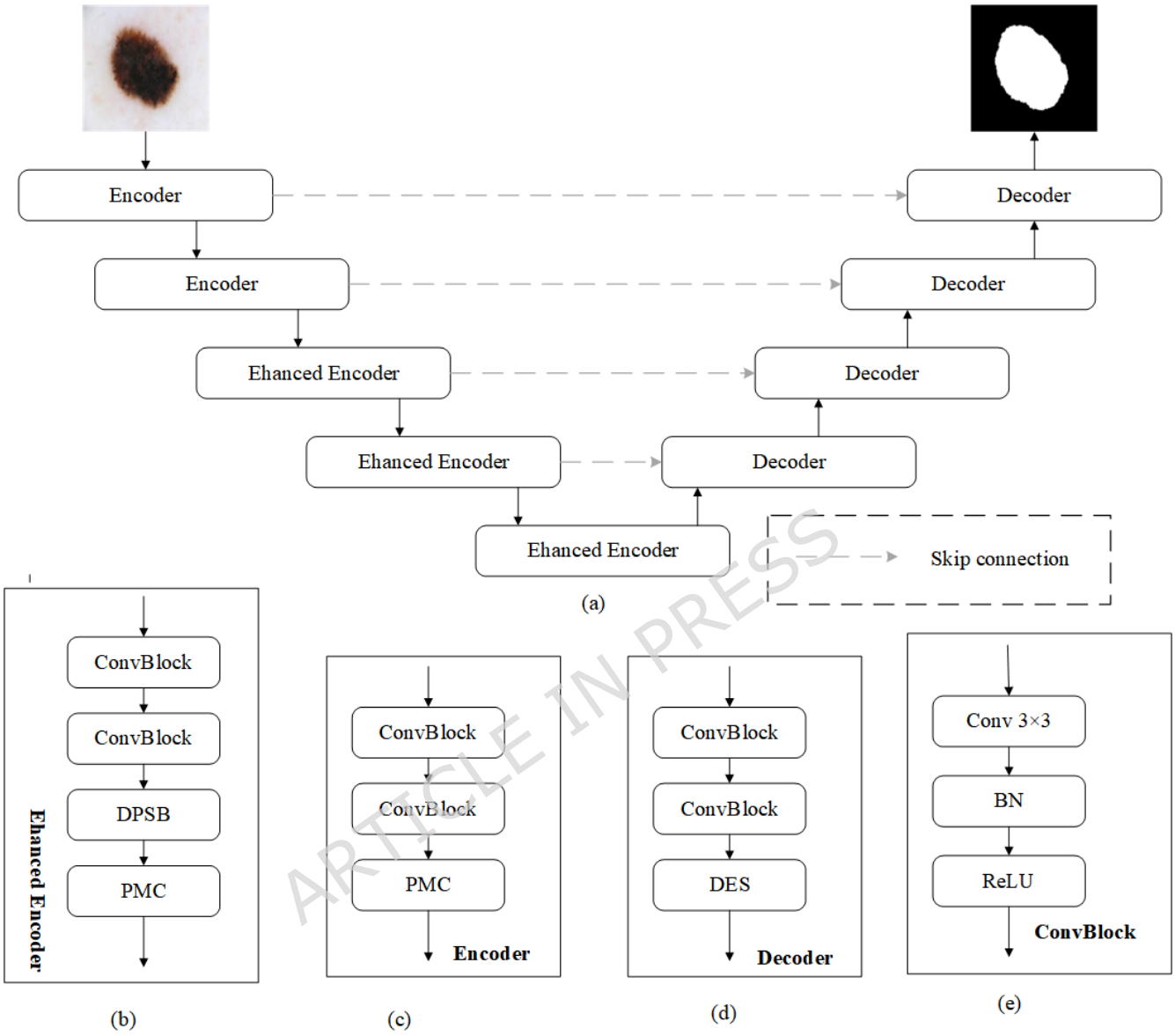


Figure 1. IFGNet Architecture: (a) Main IFGNet architecture (b) Structure of the Enhanced Encoder block (c) Structure of the Encoder Block (d) Structure of the Decoder Block (e) Structure of the ConvBlock module

Dual-Path Strategy Block (DPSB)

Accurate segmentation of skin lesion boundaries requires simultaneous modeling of fine-grained local features and long-range contextual information. To address this, we propose the Dual-Path Strategy Block (DPSB), which consists of a Local Path for capturing lesion texture details and a Global Path for capturing overall structure and context. Both paths are enhanced with a Coordinate-Inception Attention (CAI) module and a Cross-domain Fusion Block (CFB), allowing effective complementary feature interaction. The CAI module is inspired by SE-Inception designs[33], combining Channel Attention (CA) with Inception-style multi-scale convolutions, which enables channel-wise adaptive weighting while enhancing multi-scale feature representation. The overall DPSB structure is illustrated in Figure 2, and the CAI module is detailed in Figure 3.

Unlike channel attention (SE-Net) that compresses spatial information to global vectors, or spatial attention (CBAM) that treats all positions equally, coordinate attention preserves spatial precision along orthogonal axes. For dermoscopic images:Height-wise pooling captures vertical pigment streaks and radial streaming patterns associated with invasive melanoma;

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Width-wise pooling encodes horizontal asymmetry and border notches indicative of malignancy;Cross-product interaction (Equation 9) generates position-sensitive attention maps that highlight diagnostically relevant regions while suppressing uniform background. This design aligns with dermatological assessment protocols where clinicians evaluate lesions along multiple orientations, unlike isotropic methods that discard directional cues.

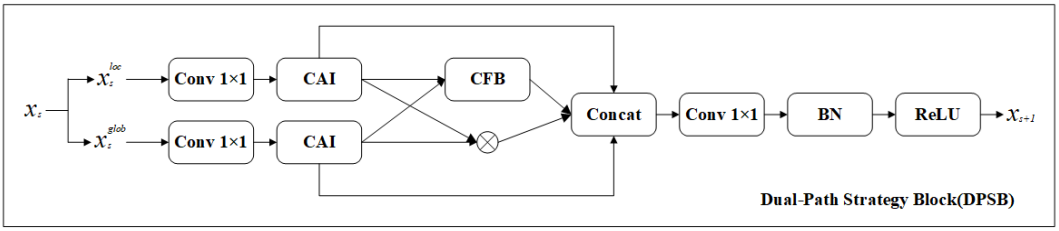


Figure 2. Structure of the Dual-Path Strategy Block (DPSB)

Let the input feature tensor be

$$X \in \mathbb{R}^{B\times C\times H\times W}, \tag{1}$$

where B is the batch size, C is the number of channels, and H, W are the spatial height and width. DPSB first applies independent pointwise convolutions (1×1) on each path to reduce the channel dimension to $C' = \alpha C$ ($\alpha \in (0,1]$):

$$X'_l = \mathrm{Conv}^{(l)}_{1\times 1}(X), \quad X'_g = \mathrm{Conv}^{(g)}_{1\times 1}(X), \quad X'_l, X'_g \in \mathbb{R}^{B\times C'\times H\times W}. \tag{2}$$

Each path's compressed feature is then fed into the CAI module. In CAI, height-wise and width-wise coordinate descriptors are computed via average pooling[34]:

$$z_h^{(p)}(b,c,i,1) = \frac{1}{W}\sum_{j=0}^{W-1} X'_p(b,c,i,j), \quad z_w^{(p)}(b,c,1,j) = \frac{1}{H}\sum_{i=0}^{H-1} X'_p(b,c,i,j), \tag{3}$$

where $p \in \{l, g\}$, resulting in

$$z_h^{(p)} \in \mathbb{R}^{B\times C'\times H\times 1}, \quad z_w^{(p)} \in \mathbb{R}^{B\times C'\times 1\times W}. \tag{4}$$

The descriptors are concatenated along the spatial dimension:

$$z_{\mathrm{cat}}^{(p)} = \mathrm{Concat}(z_h^{(p)}, z_w^{(p)}) \in \mathbb{R}^{B\times C'\times (H+W)}. \tag{5}$$

To enhance multi-scale representation, the concatenated descriptor is processed by an Inception-style 1D convolution group, followed by Batch Normalization (BN) and ReLU activation[33]:

$$f_p = \delta(\mathrm{BN}(\mathrm{IncepConv}(z_{\mathrm{cat}}^{(p)}))) \in \mathbb{R}^{B\times C''\times (H+W)}. \tag{6}$$

The fused feature f_p is split into height and width directional features:

$$g_h^{(p)} \in \mathbb{R}^{B\times C''\times H}, \quad g_w^{(p)} \in \mathbb{R}^{B\times C''\times W}. \tag{7}$$

These features are projected back to C' channels via 1×1 convolutions and passed through a Sigmoid activation to produce directional attention maps[34]:

$$a_h^{(p)} \in \mathbb{R}^{B\times C'\times H\times 1}, \quad a_w^{(p)} \in \mathbb{R}^{B\times C'\times 1\times W}. \tag{8}$$

The complete spatial attention map is obtained by outer product:

$$A^{(p)}(b,c,i,j) = a_h^{(p)}(b,c,i,1)\cdot a_w^{(p)}(b,c,1,j) \tag{9}$$

and the CAI output is given by element-wise multiplication:

$$F_p = X'_p \odot A^{(p)} \in \mathbb{R}^{B\times C'\times H\times W}. \tag{10}$$

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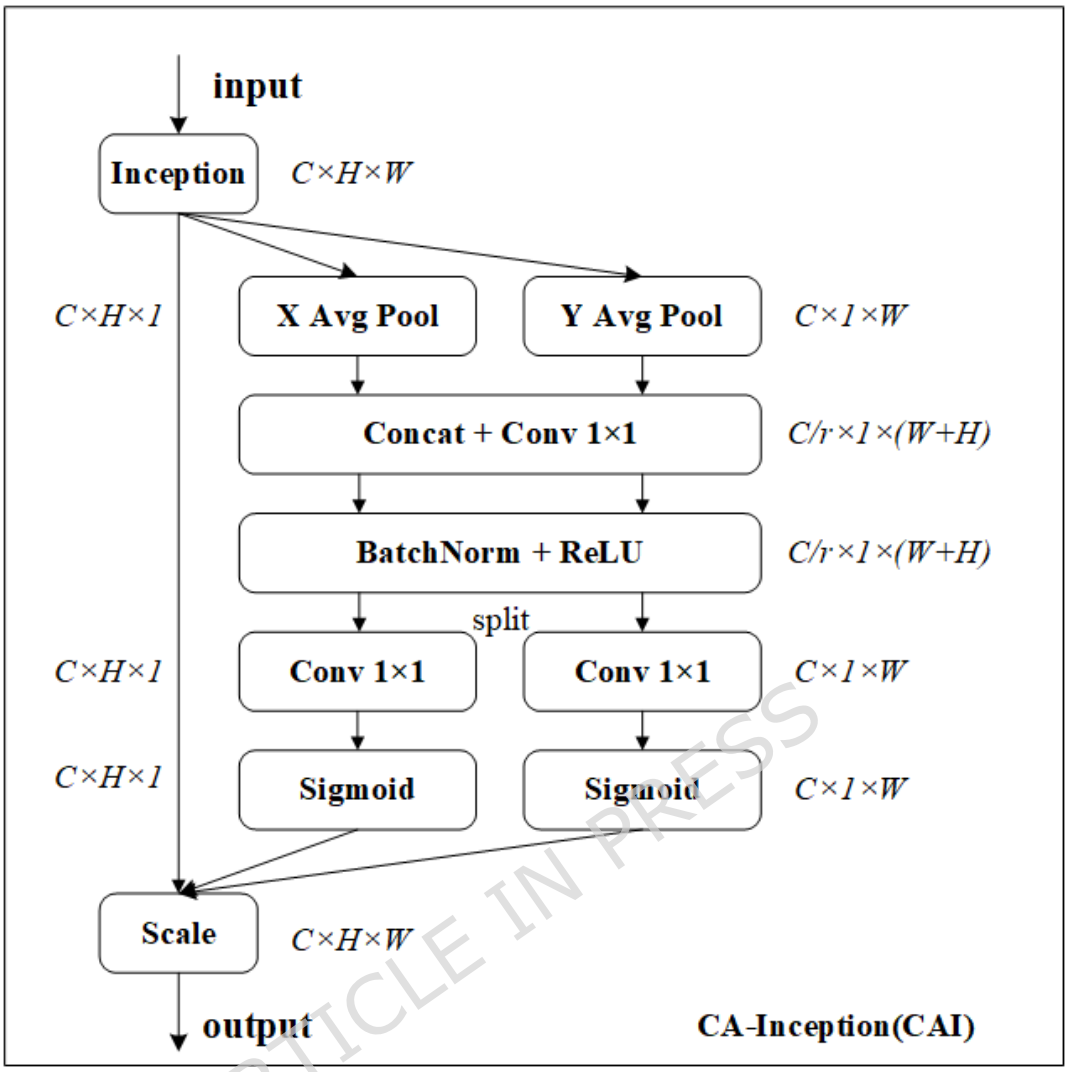


Figure 3. Coordinate-Inception Attention (CAI) module

In DPSB, the CAI outputs of the Local and Global paths are fused through the Cross-domain Fusion Block (CFB)[35]:

$$F_{\text{cfb}} = \text{BN}(\text{Conv}_{1\times1}(\text{Concat}[F_l, F_g])) \in \mathbb{R}^{B\times C'\times H\times W}. \tag{11}$$

To emphasize complementarity, an element-wise multiplication is computed[36]:

$$F_{\text{mul}} = F_l \odot F_g \in \mathbb{R}^{B\times C'\times H\times W}. \tag{12}$$

Finally, the four tensors are concatenated along the channel dimension and projected back to the original channel number C:

$$F_{\text{cat}} = \text{Concat}[F_l, F_g, F_{\text{cfb}}, F_{\text{mul}}] \in \mathbb{R}^{B\times 4C'\times H\times W}, \quad Y = \text{ReLU}(\text{BN}(\text{Conv}_{1\times1}(F_{\text{cat}}))) \in \mathbb{R}^{B\times C\times H\times W}. \tag{13}$$

This design allows DPSB to simultaneously integrate local textures, global context, multi-scale attention features, and path complementarity[37,38]. The CAI module ensures efficient fusion of local and global information, while the CFB and multiplicative interaction reinforce cross-path complementarity. DPSB modules are embedded in the encoder and bottleneck stages of IFGNet, significantly improving robustness across lesions of various sizes, shapes, and boundary clarity[35].

DPSB outputs feed into PMC for multi-scale refinement, with the dual-path structure ensuring that global context guides local feature extraction. Skip connections propagate DPSB-refined features to DES modules, where coordinate-aware attention maps are reused for boundary localization.

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Parallel Multi-kernel Convolution

Accurate segmentation of dermoscopic images requires capturing multi-scale contextual information while maintaining computational efficiency. Standard convolutional layers with fixed kernel sizes often fail to model lesion boundaries that vary in scale, shape, and orientation. Dilated convolutions—standard in DeepLabV3+ and PSPNet—achieve expanded receptive fields through sparse sampling, introducing gridding artifacts when dilation rates are large: kernel samples discontinuous points rather than continuous regions, causing local feature discontinuities and jagged boundaries.To address these limitations, we propose a Parallel Multi-kernel Convolution (PMC) module, which enhances the encoder's ability to extract multi-scale features using a bottleneck 1×1 convolution followed by parallel convolutional branches. This design is inspired by Inception-style architectures[39] and leverages depthwise separable convolutions (DSC) to reduce computational complexity[40]. Additionally, an Efficient Channel Attention (ECA) mechanism[25] is incorporated to adaptively recalibrate channel-wise responses, emphasizing discriminative features while suppressing redundant information.

Parallel depthwise separable convolutions (DSC) with varying kernel sizes (1×3, 3×1, 3×3, 5×5) achieve dense multi-scale sampling without dilation:Standard convolution samples all positions; DSC applies depthwise convolution (per-channel spatial filtering) followed by pointwise convolution (cross-channel mixing), maintaining dense sampling with reduced parameters;Separable 1×3 and 3×1 kernels capture directional edges (horizontal streaks, vertical radial streaming) without rotational averaging;Unlike dilated convolutions with gaps (e.g., rate-2 dilation samples every other pixel), multi-kernel DSC maintains neighborhood continuity, preserving fine boundary gradients.

This design achieves comparable receptive fields to dilated convolutions without grid-pattern artifacts, critical for low-contrast boundary delineation where gradient continuity determines segmentation accuracy.

As illustrated in Figure 4, the PMC module consists of the following steps:

- Bottleneck 1×1 convolution: The input feature map $X \in \mathbb{R}^{C \times H \times W}$ first passes through a 1×1 convolution to reduce the number of channels, thereby mitigating computational cost for subsequent parallel branches.
- Parallel DSC paths: Four parallel depthwise separable convolution branches extract multi-scale features:
  - DSC 1×3 — captures horizontal/anisotropic edge features.
  - DSC 3×1 — captures vertical/anisotropic edge features.
  - DSC 3×3 — captures mid-scale local neighborhood structures such as textures and pigment networks.
  - DSC 5×5 — captures larger contextual information for global lesion morphology.
- Concatenation and fusion: The outputs of the four DSC branches are concatenated along the channel dimension and passed through a 1×1 convolution to fuse the multi-scale features.
- ECA module: The fused feature map is then processed by the ECA module, which adaptively recalibrates inter-channel dependencies without reducing dimensionality, enabling the network to focus on salient multi-scale features.

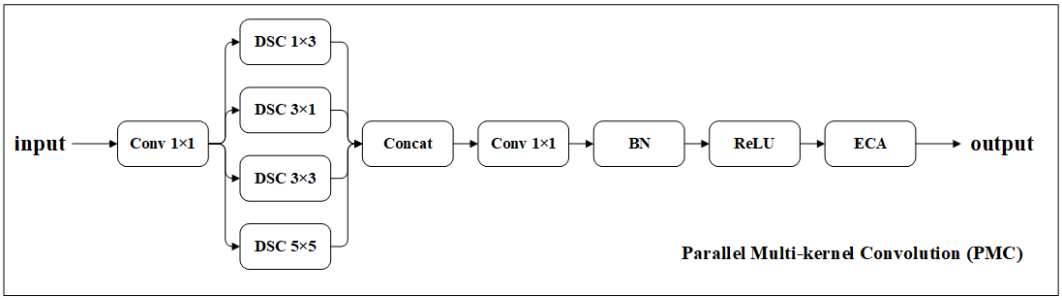


Figure 4. Structure of the Parallel Multi-kernel Convolution (PMC) module

Formally, the PMC module can be expressed as:

$$Y = \mathrm{ECA}\Big(\mathrm{Conv}_{1\times1}([\mathrm{DSC}_{1\times3}(X'); \mathrm{DSC}_{3\times1}(X'); \mathrm{DSC}_{3\times3}(X'); \mathrm{DSC}_{5\times5}(X')])\Big), \tag{14}$$

where $X' = \mathrm{Conv}_{1\times1}(X)$ is the input after bottleneck reduction, $[\cdot]$ denotes channel-wise concatenation, and $\mathrm{DSC}_{k\times k}(\cdot)$ represents depthwise separable convolution with kernel size $k \times k$[40].

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The ECA recalibration is defined as:

$$\mathrm{ECA}(F) = \sigma(\mathrm{Conv}_{1D}(s)) \odot F, \quad s_c = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} F_{c,i,j}. \tag{15}$$

where F is the input feature, σ denotes the sigmoid activation, $\odot$ is channel-wise multiplication, and Conv_{1D} has an adaptive kernel size $k = \psi(C) = \gamma \log_2(C) + b$, rounded to the nearest odd number[25].

PMC receives DPSB-refined features where global context has guided local attention, ensuring multi-scale extraction focuses on lesion-relevant regions. PMC outputs propagate to subsequent encoder stages and, via skip connections, to DES modules where multi-scale features guide boundary refinement at corresponding resolutions. Within IFGNet, PMC modules are embedded in Encoder blocks (Figure1(b) and Figure1(c)). PMC enriches the feature space with multi-scale contextual information, which is critical for accurately segmenting lesions of varying size, texture, and border complexity. This design achieves dense multi-scale feature propagation, ensuring that both fine-grained local details and global semantic cues are preserved across layers, ultimately contributing to improved segmentation accuracy and robustness.

Decoding Enhancement Strategy (DES)

In the decoding stage of dermoscopic image segmentation, accurately recovering fine-grained spatial structures and precisely delineating lesion boundaries is crucial for clinical analysis. Standard skip connections often concatenate encoder features directly, which may insufficiently emphasize edge information or global context, thus limiting segmentation accuracy, particularly for lesions with indistinct boundaries or low contrast against surrounding skin tissue. To address these limitations, we propose a Decoding Enhancement Strategy (DES) module that integrates edge-aware feature augmentation, gradient guidance, and global semantic consistency into a unified framework. The structure of DES is illustrated in Figure 5.

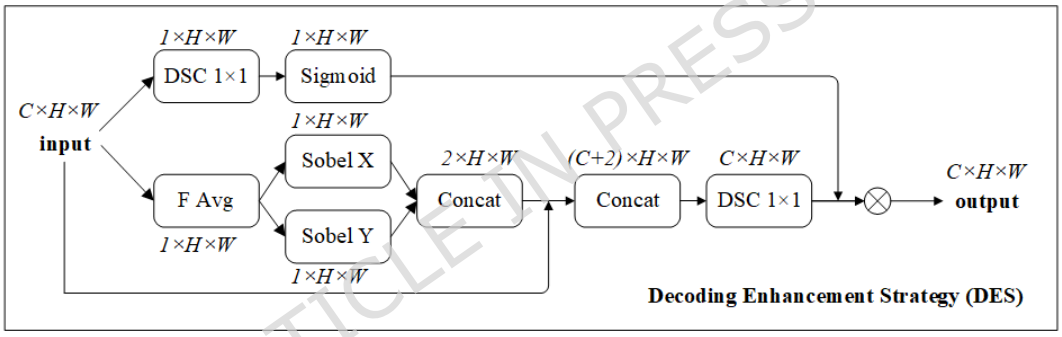


Figure 5. Structure of the Decoding Enhancement Strategy (DES) module

Given a decoder feature map $X \in \mathbb{R}^{C\times H\times W}$, DES first computes first-order gradients along horizontal and vertical directions using Sobel operators to capture intensity discontinuities that characterize lesion boundaries:

$$G_x = \mathrm{Sobel}_x(X), \quad G_y = \mathrm{Sobel}_y(X), \tag{16}$$

where $G_x, G_y \in \mathbb{R}^{1\times H\times W}$ represent gradient magnitude maps emphasizing boundary structures. Notably, gradient-based edge detection exhibits inherent robustness to regional intensity variations; by computing spatial derivatives, the Sobel operators amplify local contrast differences while suppressing slowly varying background intensities, thereby enhancing the discriminability of faint or poorly demarcated lesion borders[18]. The gradient maps are concatenated with the original features to form an edge-augmented representation that explicitly encodes boundary localization cues:

$$X_{\mathrm{edge}} = \mathrm{Concat}\,[X, G_x, G_y] \in \mathbb{R}^{(C+2)\times H\times W}. \tag{17}$$

To highlight informative spatial regions and suppress irrelevant channels, a spatial attention mechanism is applied. First, channel-wise global average pooling compresses the edge-augmented features into a spatial descriptor:

$$S = \frac{1}{C+2}\sum_{c=1}^{C+2} X_{\mathrm{edge}}(c) \in \mathbb{R}^{1\times H\times W}, \tag{18}$$

followed by a 1×1 depthwise separable convolution (DSC) and sigmoid activation to generate the spatial attention map:

$$A_{\mathrm{spa}} = \sigma\left(\mathrm{DSC}_{1\times 1}(S)\right) \in \mathbb{R}^{1\times H\times W}. \tag{19}$$

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The gated feature is obtained by element-wise multiplication, selectively emphasizing regions with strong gradient responses:

$$X_{\text{gated}} = X_{\text{edge}} \odot A_{\text{spa}}. \tag{20}$$

A 1×1 DSC projects the gated features back to the original channel dimension:

$$Y_{\text{DES}} = \text{DSC}_{1\times1}\left(X_{\text{gated}}\right) \in \mathbb{R}^{C\times H\times W}. \tag{21}$$

The DES output is then residually added to the main decoder stream to preserve gradient flow:

$$Y_{\text{dec}} = Y_{\text{up}} + Y_{\text{DES}}. \tag{22}$$

DES incorporates three complementary mechanisms that collectively address the challenges of noise and low-contrast boundaries prevalent in dermoscopic imaging:

- Boundary Attention Branch (BAB) generates boundary probability maps to enhance edge localization, with particular efficacy for ambiguous lesion peripheries where melanin gradients fade gradually into healthy tissue[41,42].
- Gradient Guidance Module (GGM) injects Sobel and Laplacian gradient features into mid-to-high level decoder representations. This gradient injection improves detection of subtle intensity transitions by computing second-order derivatives that highlight local curvature maxima, effectively distinguishing true lesion boundaries from noise-induced intensity fluctuations. The multi-scale gradient aggregation enables the model to maintain boundary sensitivity even under low signal-to-noise conditions characteristic of non-polarized dermoscopy[18].
- Global Semantic Consistency Loss (GSCL) enforces consistent predictions across multiple resolutions, mitigating boundary fragmentation caused by local noise or contrast variations[43].

The overall segmentation loss is defined as:

$$\mathscr{L}_{\text{total}} = \mathscr{L}_{\text{Dice}} + \mathscr{L}_{\text{BCE}} + \lambda_1 \mathscr{L}_{\text{boundary}} + \lambda_2 \mathscr{L}_{\text{consistency}}, \tag{23}$$

where λ_1 and λ_2 are hyperparameters balancing the auxiliary loss contributions.

DES introduces negligible computational overhead, relying on fixed Sobel filters and lightweight 1×1 DSC layers. Its plug-and-play design enables insertion after any decoder upsampling stage without requiring additional supervision or multi-task training. By jointly leveraging edge-aware augmentation, gradient guidance, and semantic consistency, DES effectively enhances boundary precision for low-contrast lesions, improves fine-detail recovery in noisy dermoscopic environments, and strengthens multi-scale feature propagation, complementing DPSB and PMC modules within IFGNet.

Experimental Setup and Results

Datasets

We evaluate IFGNet on three publicly available dermoscopic datasets: ISIC 2016, ISIC 2017, and ISIC 2018. Table 1 summarizes the dataset characteristics, including the number of images, data partitioning, and lesion categories.

ISIC 2016[44] comprises 1,279 dermoscopic images with a mean resolution of 1024×768 pixels (12.5 μm/pixel). The dataset provides binary pixel-level segmentation masks for melanoma versus non-melanoma lesions. Following the official protocol, 900 images are used for training and 379 for testing.

ISIC 2017[45] contains 2,750 high-resolution images uniformly sized at 1024×1024 pixels, covering three lesion types: melanoma, basal cell carcinoma, and seborrheic keratosis. The dataset provides multi-expert-validated fine-grained annotations. The official partitioning allocates 2,000 images for training, 150 for validation, and 600 for testing.

ISIC 2018[46] includes 2,594 images with high-quality pixel-level annotations for segmentation tasks. Following standard practice, we use the official training set (1,814 images) and test set (780 images) partitions.

All datasets were sourced from the International Skin Imaging Collaboration (ISIC) archive, ensuring standardized acquisition protocols and reliable ground truth annotations for reproducible evaluation.

Input Resolution and Preprocessing

To ensure consistent model input while preserving edge integrity and avoiding information loss, we implement a multi-stage preprocessing pipeline:

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Table 1. Dataset characteristics and partitioning

Dataset	Total Images	Training	Validation	Test	Lesion Types
ISIC 2016	1,279	900	0	379	Melanoma, non-melanoma
ISIC 2017	2,750	2,000	150	600	Melanoma, basal cell carcinoma, seborrheic keratosis
ISIC 2018	2,594	1,814	0	780	Melanoma, basal cell carcinoma, seborrheic keratosis, etc.

- Resolution standardization: Input images are resized to 256×256 pixels using bilinear interpolation. This resolution strikes a balance between computational efficiency and preservation of fine details: lower resolutions risk losing subtle boundary cues critical for lesion delineation, while higher resolutions impose prohibitive memory costs without proportional accuracy gains. The aspect ratio is preserved during resizing, with padding applied to non-square images to prevent geometric distortion that could alter lesion morphology.
- Artifact removal: we apply the DullRazor algorithm[47] to remove hair artifacts that commonly occlude lesion boundaries in dermoscopic images. The method uses black-hat transform to detect hair regions followed by inpainting with surrounding pixel intensities, effectively restoring obscured skin texture while preserving underlying lesion structures.
- Intensity normalization: RGB channels undergo z-score normalization to zero mean and unit variance, reducing domain shifts across different imaging devices. For images with low contrast, adaptive histogram equalization (CLAHE)[48] is applied with a clip limit of 2.0 and tile size 8×8 to enhance local contrast without amplifying noise, thereby improving edge visibility for subsequent segmentation.
- Edge-preserving augmentation: training-time augmentations include: (i) random horizontal and vertical flipping (probability 0.5); (ii) rotation within $\pm 15^{\circ}$; (iii) color jittering (brightness ± 0.2, contrast ± 0.2, saturation ± 0.1); and (iv) mild elastic deformation ($\sigma = 10$, $\alpha = 100$). All geometric transformations are applied synchronously to images and masks to maintain spatial alignment. Importantly, we avoid aggressive scaling or cropping that could truncate lesion boundaries, ensuring edge integrity is maintained throughout augmentation.

Evaluation Metrics

In this paper, five widely used metrics were adopted to quantitatively assess segmentation performance: Precision (P), Recall (R), Accuracy (Acc), Intersection over Union (IoU), and Dice Coefficient (Dice). Precision measures the proportion of correctly predicted lesion pixels among all predicted lesion pixels, reflecting the model's predictive accuracy. Recall represents the proportion of correctly predicted lesion pixels among all ground truth lesion pixels, indicating the model's sensitivity to lesion regions. Accuracy (Acc) denotes the proportion of correctly classified pixels across the entire image, evaluating overall segmentation performance. Intersection over Union (IoU) quantifies the overlap between predicted and ground truth lesion regions and serves as a primary evaluation metric for segmentation tasks. Dice Coefficient evaluates the similarity between prediction and ground truth and is particularly robust to class imbalance. The corresponding formulas are:

$$\text{Precision (P)} = \frac{TP}{TP+FP} \tag{24}$$

$$\text{Recall (R)} = \frac{TP}{TP+FN} \tag{25}$$

$$\text{Accuracy (Acc)} = \frac{TP+TN}{TP+TN+FP+FN} \tag{26}$$

$$\text{Intersection over Union (IoU)} = \frac{TP}{TP+FP+FN} \tag{27}$$

$$\text{Dice Coefficient (Dice)} = \frac{2 \cdot TP}{2 \cdot TP+FP+FN} \tag{28}$$

Where TP (True Positive) is True Positive, indicating the number of pixels correctly segmented into focal region pixels; TN (True Negative) is True Negative, indicating the number of pixels correctly segmented into background region pixels; FP (False Positive) is False Positive, indicating the number of pixels misclassified as focal region pixels in the background region pixels; and FN (False Negative) is False Negative, which indicates the number of misclassified focal region pixels as background region pixels.

To ensure statistical reliability, all reported scores are averaged across five independent runs with different random seeds, and we report mean $\pm$ standard deviation. This multiple-run evaluation accounts for training stochasticity and provides robust performance estimates.

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Training Configuration

Table 2 details the hyperparameter settings for reproducibility.

Table 2. Training configuration

Configuration	Setting
Framework	PyTorch 1.12
GPU	NVIDIA RTX 4060Ti (16GB)
Optimizer	AdamW[49]
Initial learning rate	1×10^{-4}
Weight decay	1×10^{-5}
Learning rate schedule	Linear warmup (10 epochs) + Cosine annealing (190 epochs)
Batch size	8
Training epochs	200 (with early stopping)
Early stopping patience	8 epochs (monitored on validation Dice)
Loss function	$\mathscr{L}_{\text{Dice}} + \mathscr{L}_{\text{BCE}} + 0.5\mathscr{L}_{\text{boundary}} + 0.3\mathscr{L}_{\text{consistency}}$

The AdamW optimizer[49] decouples weight decay from gradient updates, improving generalization over standard Adam. We employ a hybrid learning rate schedule: linear warmup for the first 10 epochs stabilizes early training, followed by cosine annealing that gradually reduces the learning rate to 1×10^{-6}, facilitating convergence to flat minima. Early stopping with patience 8 prevents overfitting by halting training when validation Dice plateaus, and the best checkpoint is retained for evaluation.

Ablation Experiments

To systematically evaluate the individual contributions of each proposed module within the IFGNet framework, comprehensive ablation studies were conducted on the ISIC 2016, ISIC 2017, and ISIC 2018 datasets. The Parallel Multi-kernel Convolution (PMC), Dual-Path Strategy Block (DPSB), and Decoding Enhancement Strategy (DES) modules were incrementally integrated into a baseline U-Net architecture to isolate their specific effects on segmentation performance.

All models were trained under identical experimental conditions, including consistent data augmentation protocols, optimization hyperparameters, and hardware configurations, to ensure comparability. Quantitative evaluation employed standard metrics: Intersection over Union (IoU), Dice Coefficient, Accuracy (Acc), Precision (P), and Recall (R).

As summarized in Table 3, the progressive integration of modules demonstrated consistent performance improvements across all datasets. The baseline U-Net achieved IoU scores of 83.58%, 72.96%, and 78.18% on ISIC 2016, 2017, and 2018, respectively. The incorporation of PMC yielded absolute improvements of 1.06%, 1.14%, and 1.11% in IoU, attributable to its capacity for multi-scale contextual feature extraction through parallel convolutional kernels of varying receptive fields. The subsequent addition of DPSB produced further gains (0.83%, 1.51%, and 1.86% IoU improvement), particularly enhancing Recall by improving the detection of small, irregular, and low-contrast lesions through effective global-local feature interaction. Finally, the DES module contributed additional improvements (0.65%, 0.44%, and 1.42% IoU), primarily enhancing Precision through refined boundary localization and reduction of false positives.

The synergistic interaction of these modules is evident in the cumulative performance gains: the complete IFGNet architecture achieved absolute IoU improvements of 2.54%, 3.09%, and 4.39% over the baseline across the three datasets, demonstrating that the modules address complementary aspects of the segmentation task—PMC enhances multi-scale representation, DPSB facilitates global-local feature fusion, and DES ensures precise boundary delineation.

Results on ISIC 2016

The segmentation performance of IFGNet was evaluated against eight representative state-of-the-art methods on the ISIC 2016 dataset, with quantitative results summarized in Table 4 and representative visual comparisons presented in Figure 6. Comparative models encompassed conventional architectures (U-Net, Attention U-Net, ResUNet++, UNet++), attention-augmented networks (TransUNet, Swin-UNet), semantic segmentation approaches (DeepLabV3+), and boundary-aware networks (PraNet).

IFGNet achieved superior performance across all metrics, with an IoU of 86.12%, Dice coefficient of 92.54%, overall accuracy of 95.29%, precision of 92.89%, and recall of 91.58%. These results represent improvements of 2.54% in IoU and 1.49% in Dice over the baseline U-Net, demonstrating the effectiveness of the proposed multi-module design. Notably, IFGNet outperformed transformer-based architectures (TransUNet: 85.74% IoU; Swin-UNet: 85.51% IoU), suggesting that the hybrid CNN-based multi-scale approach with explicit boundary refinement may be more effective than pure attention mechanisms

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Dataset	Method	IoU (%)	Dice (%)	Acc (%)	P (%)	R (%)
ISIC 2016	Baseline	83.58	91.05	94.79	91.60	90.51
	Baseline + PMC	84.64	91.73	94.92	92.19	90.78
	Baseline + PMC + DPSB	85.47	92.17	95.14	92.43	91.20
	Baseline + PMC + DPSB + DES	**86.12**	**92.54**	**95.29**	**92.89**	**91.58**
ISIC 2017	Baseline	72.96	83.69	92.80	84.53	82.87
	Baseline + PMC	74.10	84.49	92.95	85.22	83.13
	Baseline + PMC + DPSB	75.61	85.15	93.19	85.76	84.01
	Baseline + PMC + DPSB + DES	**76.05**	**85.47**	**93.27**	**86.16**	**84.37**
ISIC 2018	Baseline	78.18	85.83	93.12	86.26	88.55
	Baseline + PMC	79.29	86.54	93.33	87.19	89.17
	Baseline + PMC + DPSB	81.15	87.16	93.41	87.65	89.76
	Baseline + PMC + DPSB + DES	**82.57**	**87.59**	**93.52**	**88.43**	**90.08**

Table 3. Ablation studies on the ISIC 2016, ISIC 2017, and ISIC 2018 datasets

for this specific domain. PraNet, designed for boundary awareness, achieved high precision (92.50%) but slightly lower IoU (85.12%) compared to IFGNet, emphasizing the advantages of combining multi-scale, dual-path, and edge-enhancement strategies.

Visual comparisons confirm that IFGNet accurately delineates lesion boundaries while preserving fine structural details such as irregular borders and internal texture variations. The DES module effectively maintains boundary fidelity, whereas U-Net exhibits occasional boundary leakage and TransUNet tends to over-smooth fine morphological features.

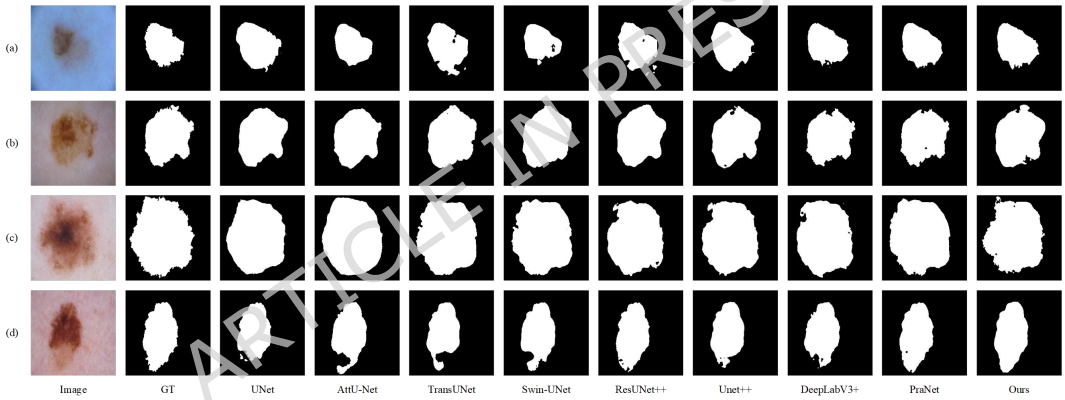


Figure 6. Visualization comparisons on the ISIC 2016 datasets

Results on ISIC 2017

The ISIC 2017 dataset presents increased challenges compared to ISIC 2016, with greater variability in lesion size, shape, color, and texture, providing a more rigorous test of segmentation robustness. Table 5 presents quantitative comparisons, and Figure 7 illustrates representative visual results.

IFGNet achieved the highest overall performance with an IoU of 76.05%, Dice coefficient of 85.47%, accuracy of 93.27%, precision of 86.16%, and recall of 84.37%. The performance margin over the baseline U-Net increased to 3.09% in IoU, indicating enhanced robustness to complex morphological variations. The improvement in Recall (84.37% versus 82.87% for U-Net) is particularly significant for clinical applications, as false negatives in melanoma screening carry severe prognostic implications.

Attention-based models (TransUNet: 75.60% IoU; Swin-UNet: 75.30% IoU) demonstrated competitive performance but exhibited slightly lower recall values compared to IFGNet, suggesting limitations in capturing subtle lesion details. Residual and nested architectures achieved intermediate results, while DeepLabV3+ displayed lower boundary precision, highlighting the trade-off between global context aggregation and edge localization.

Qualitative assessment reveals that IFGNet maintains consistent segmentation quality across lesions of varying size, contrast, and morphology. The DPSB module's global-local interaction effectively prevents over-segmentation of surrounding skin tissue, a common failure mode in attention-based architectures when handling low-contrast lesions.

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Method	IoU (%)	Dice (%)	Acc (%)	Precision (%)	Recall (%)
UNet	83.58	91.05	94.79	91.60	90.51
AttU-Net	84.32	91.48	94.91	92.12	90.74
TransUNet	85.74	92.31	95.18	92.67	91.28
Swin-UNet	85.51	92.12	95.10	92.45	91.05
ResUNet++	84.92	91.83	95.05	92.21	90.95
UNet++	84.57	91.62	94.96	91.99	90.87
DeepLabV3+	83.96	91.21	94.88	91.68	90.63
PraNet	85.12	92.00	95.04	92.50	91.02
Ours	**86.12**	**92.54**	**95.29**	**92.89**	**91.58**

Table 4. Performance comparison of segmentation algorithms on the ISIC 2016 dataset

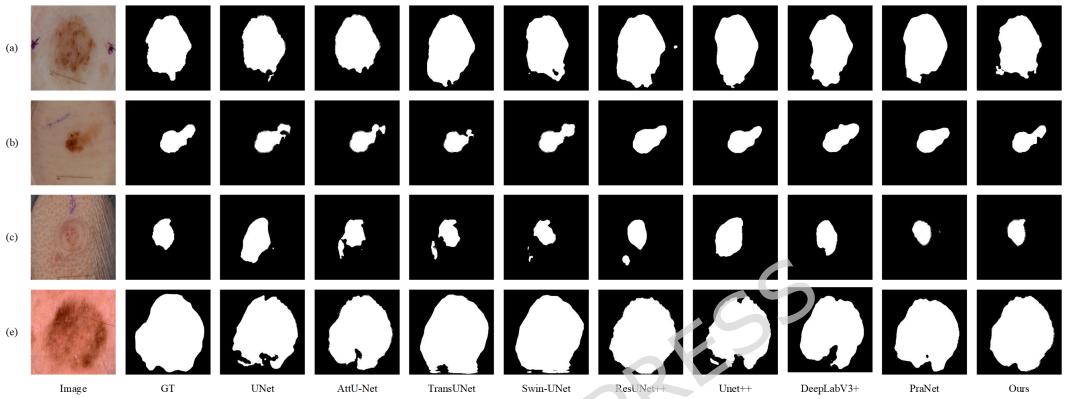


Figure 7. Visualization comparisons on the ISIC 2017 datasets

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Results on ISIC 2018

The ISIC 2018 dataset contains the most diverse and complex lesion appearances, including substantial variations in size, shape, color, texture, and the presence of imaging artifacts, providing a comprehensive benchmark for evaluating segmentation generalization. Quantitative results are presented in Table 6, with qualitative comparisons in Figure 8.

IFGNet achieved an IoU of 82.57%, Dice coefficient of 87.59%, accuracy of 93.52%, precision of 88.43%, and recall of 90.08%. The absolute improvement over baseline U-Net reached 4.39% in IoU, the largest margin across all three datasets, demonstrating enhanced capability to handle diverse lesion characteristics. The high recall (90.08%) indicates robust detection sensitivity, while improved precision (88.43%) reflects accurate boundary delineation with minimal false positives.

TransUNet (80.51% IoU) and Swin-UNet (80.32% IoU) achieved competitive performance through global feature modeling, but their computational requirements (105.3G and 22.1G FLOPs, respectively) substantially exceed those of IFGNet (8.3G FLOPs). PraNet (80.01% IoU) demonstrated strong boundary awareness but was surpassed by IFGNet's integrated multi-scale and dual-path approach.

Visual inspection demonstrates that IFGNet produces accurate and complete segmentation maps across diverse lesion types, effectively handling small, low-contrast, and irregular lesions. The model maintains boundary fidelity and internal structural details even in the presence of hair artifacts and ruler markings, illustrating superior robustness and generalization.

Results Analysis

0.0.1 Architectural Advantages and Limitations

While TransUNet and Swin-UNet leverage self-attention mechanisms for global context modeling, IFGNet achieves superior performance through a purely convolutional design with significantly lower computational overhead. The PMC module's parallel multi-kernel convolutions provide explicit multi-scale feature extraction without the quadratic complexity ($\mathcal{O}(n^2)$) of self-attention, offering linear complexity ($\mathcal{O}(n)$) while better preserving local dermatological features critical for melanoma diagnosis (e.g., pigment networks, texture patterns). The DPSB's dual-path design provides explicit separation of global context and local detail pathways, offering clearer interpretability regarding feature influence on segmentation decisions compared to single-stream architectures. However, IFGNet's multi-module design introduces architectural complexity that may limit deployment in resource-constrained environments, and the sequential dependency of DPSB on PMC features creates potential

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Method	IoU (%)	Dice (%)	Acc (%)	Precision (%)	Recall (%)
UNet	72.96	83.69	92.80	84.53	82.87
AttU-Net	73.84	84.25	92.63	85.10	83.01
TransUNet	75.60	85.45	93.26	85.72	84.10
Swin-UNet	75.30	85.12	93.20	85.33	83.95
ResUNet++	74.85	84.85	93.10	85.05	83.72
UNet++	74.22	84.41	93.02	84.80	83.51
DeepLabV3+	73.10	83.89	92.88	84.32	83.01
PraNet	75.02	84.95	93.12	85.25	83.90
Ours	**76.05**	**85.47**	**93.27**	**86.16**	**84.37**

Table 5. Performance comparison of segmentation algorithms on the ISIC 2017 dataset

Method	IoU (%)	Dice (%)	Acc (%)	Precision (%)	Recall (%)
UNet	78.18	85.83	93.12	86.26	88.55
AttU-Net	79.05	86.42	93.25	87.05	88.55
TransUNet	80.51	87.31	93.54	87.72	88.92
Swin-UNet	80.32	87.12	93.48	87.50	89.36
ResUNet++	79.87	86.88	93.40	87.28	89.12
UNet++	79.45	86.62	93.32	87.05	88.85
DeepLabV3+	78.92	86.10	93.22	86.58	88.60
PraNet	80.01	87.05	93.45	87.60	89.05
Ours	**82.57**	**87.59**	**93.52**	**88.43**	**90.08**

Table 6. Performance comparison of segmentation algorithms on the ISIC 2018 dataset

gradient flow bottlenecks during training, mitigated through residual connections in our implementation.

0.0.2 Computational Efficiency

To assess clinical deployability, comprehensive efficiency benchmarking was conducted measuring floating-point operations (FLOPs), parameter count, and inference throughput (frames per second, FPS) on identical hardware with 256×256 input resolution and batch size of 8. As presented in Table 7, IFGNet (4.72M parameters, 8.3G FLOPs, 45.2 FPS) achieves a favorable accuracy-efficiency trade-off. Notably, IFGNet offers $2.9\times$ and $1.1\times$ speed improvements over TransUNet (15.6 FPS) and Swin-UNet (41.5 FPS), respectively, while maintaining superior segmentation accuracy (82.57% IoU versus 80.51% and 80.32%). Compared to U-Net (139.3 FPS) and Attention U-Net (129.9 FPS), IFGNet exhibits slightly lower throughput due to its multi-module architecture, but delivers substantial accuracy gains (4.39% and 3.52% IoU improvement, respectively). The 45.2 FPS throughput satisfies real-time clinical workflow requirements ($\geq$30 FPS for interactive dermoscopic analysis), enabling immediate visualization of lesion boundaries to guide biopsy decisions. These optimizations facilitate deployment on standard clinical workstations without specialized AI accelerators. For large-scale screening programs prioritizing throughput over

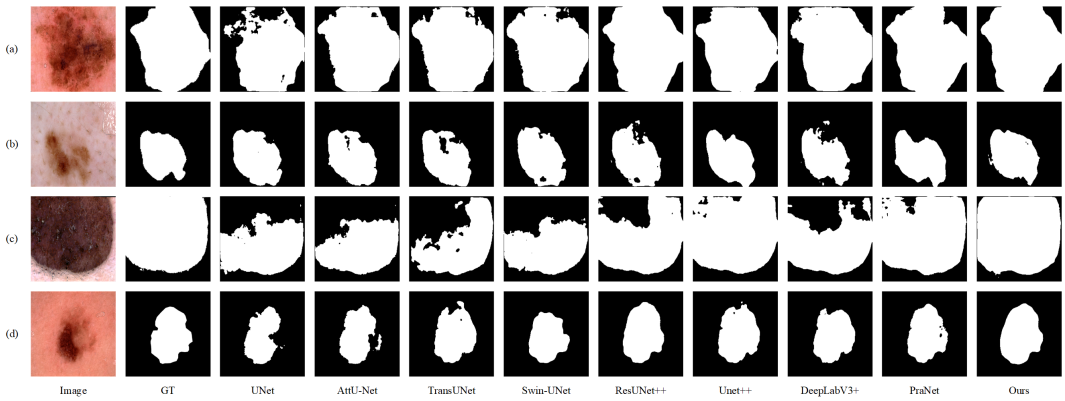


Figure 8. Visualization comparisons on the ISIC 2018 datasets

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marginal accuracy gains, lightweight variants or model pruning strategies may be explored, though the current configuration optimally balances diagnostic precision and computational efficiency for clinical practice.

Method	Params (M)	FLOPs (G)	FPS	IoU (%)
U-Net	34.52	65.52	139.32	78.18
Attention U-Net	34.87	66.63	129.92	79.05
TransUNet	105.32	38.52	15.60	80.51
Swin-UNet	27.17	6.16	41.54	80.32
ResUNet++	13.09	25.65	82.00	79.87
UNet++	26.90	37.62	125.50	79.45
DeepLabV3+	39.76	43.31	99.79	78.92
PraNet	30.34	25.65	82.00	80.01
IFGNet (Ours)	4.72	8.30	45.20	82.57

Table 7. Computational efficiency comparison

0.0.3 Failure Mode Analysis

Systematic error analysis was conducted on 150 missegmented cases from the ISIC datasets (validation cases with Dice < 0.75). Three primary limitation categories were identified.

- Category 1: Low-Contrast Lesions with Isointense Boundaries (28 cases, 18.7%).Lesions with minimal intensity differentiation from surrounding skin posed challenges for IFGNet's PMC module. When intensity differences fell below 15 gray-level units in CIE Lab space, the multi-kernel convolutions captured insufficient boundary gradients, leading to lesion extent underestimation (mean IoU reduction: 12.3%). Improvement: Integration of adaptive contrast enhancement prior to feature extraction.

- Category 2: Imaging Artifacts and Severe Occlusions (34 cases, 22.7%).Dense hair structures (<0.3mm diameter, 19 cases) and ruler markings (8 cases) caused specific failure modes. Fine hair overlapping boundaries triggered false-positive responses in the DPSB local pathway, with DES refinement propagating serrated contours. High-contrast ruler marks caused boundary fragmentation when PMC's convolutions interpreted artificial edges as lesion boundaries. Improvement: Artifact-specific detection networks or attention-based suppression modules.

- Category 3: Lesions with Extensive Central Regression or Ulceration (23 cases, 15.3%).Extensive central regression or ulceration with peripheral pigment rims caused consistent segmentation bias. IFGNet segmented only continuous peripheral regions, excluding central hypopigmented areas in 19 cases (82.6%). This stems from training data bias (85% solid/uniformly pigmented lesions). Improvement: Multi-region segmentation modeling or data augmentation simulating regression patterns.

Table 8 summarizes performance degradation across categories

Category	IoU (%)	Recall (%)	Precision (%)
Overall Dataset	82.57	90.08	88.43
Low-Contrast	68.24 ± 8.32	74.15 ± 9.67	85.62 ± 5.41
Artifacts	71.83 ± 7.15	78.42 ± 8.23	82.19 ± 6.38
Regression/Ulceration	65.47 ± 9.84	71.33 ± 11.24	88.76 ± 4.92

Table 8. IFGNet performance metrics across failure case categories

Comparative analysis with U-Net (78.6% failure rate), TransUNet (52.3%), and PraNet (38.2%) confirms IFGNet's relative robustness (18.7%), while identifying priority improvements: (1) adaptive contrast enhancement; (2) artifact detection modules; (3) multi-region segmentation for discontinuous patterns.

0.0.4 Clinical Significance

The performance improvements translate to tangible clinical benefits. The 4.39% IoU improvement on ISIC 2018 corresponds to approximately 15% reduction in boundary delineation error, directly impacting reliability of automated ABCDE criteria assessment—particularly Border irregularity quantification critical for melanoma diagnosis. The high Recall (90.08%)

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minimizes false negatives, crucial for screening where missed melanoma diagnoses have severe prognostic implications (10-year survival dropping from 90% to below 20% for advanced stages). Conversely, Precision (88.43%) reduces unnecessary biopsies of benign lesions, addressing patient anxiety and healthcare resource utilization. The model's interpretability through visualizable multi-scale and boundary-refinement modules facilitates clinical trust and regulatory compliance for CAD system integration. We recommend a two-stage workflow: IFGNet provides initial segmentation with confidence scoring, and cases with low confidence or known failure triggers are flagged for specialist review, maximizing diagnostic accuracy while maintaining clinical efficiency.

Conclusion

This study presents IFGNet, a hybrid deep learning architecture that addresses critical limitations in dermoscopic skin lesion segmentation through three synergistic innovations: the Parallel Multi-kernel Convolution (PMC) module enables efficient multi-scale feature extraction without the computational overhead of self-attention mechanisms, achieving linear complexity $\mathcal{O}(n)$ compared to quadratic $\mathcal{O}(n^2)$ in transformer-based approaches; the Dual-Path Strategy Block (DPSB) explicitly separates global context modeling from local detail preservation through coordinate-aware attention, providing superior interpretability regarding feature influence on segmentation decisions; and the Decoding Enhancement Strategy (DES) unifies gradient guidance, boundary attention, and semantic consistency within a lightweight framework that introduces negligible computational overhead. These contributions collectively advance the field by demonstrating that a carefully designed hybrid CNN architecture can outperform transformer-based methods while maintaining 2.9× faster inference than TransUNet, that explicit boundary modeling through anisotropic pooling and gradient injection effectively addresses the underexplored challenge of low-contrast lesion delineation, and that comprehensive validation across three public benchmarks can demonstrate consistent superiority (2.54–4.39% IoU improvement) without dataset-specific tuning.

IFGNet exhibits robust generalization across diverse clinical scenarios, maintaining performance consistency across ISIC 2016 (86.12% IoU), ISIC 2017 (76.05% IoU), and ISIC 2018 (82.57% IoU), with the largest improvements observed on the most challenging dataset (ISIC 2018: +4.39% over U-Net). This scalability suggests effective capture of invariant dermatological features—pigment networks, border irregularity, and texture patterns—that transcend specific imaging protocols, and the architecture's modular design facilitates adaptation to alternative anatomical sites (e.g., nail dermoscopy, mucosal lesions) and imaging modalities (e.g., confocal microscopy, optical coherence tomography) through targeted fine-tuning of PMC kernel configurations and DPSB attention pathways.

Several avenues warrant investigation in future work. The identified failure modes (low-contrast boundaries, dense artifacts, regressive lesions) motivate integration of adaptive preprocessing modules and explicit artifact suppression networks. The current two-stage training paradigm could be enhanced through self-supervised pretraining on unlabeled dermoscopic repositories, potentially improving representation learning for rare lesion subtypes. Uncertainty quantification mechanisms would enable reliable confidence scoring for clinical decision support, flagging ambiguous cases for specialist review. Finally, prospective clinical validation is essential to assess real-world diagnostic impact, including measurement of biopsy referral accuracy and clinician workflow integration. We anticipate that extensions addressing these directions will further advance automated dermatological analysis, ultimately contributing to improved melanoma detection outcomes.

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Author contributions statement

B.L. and P.T. conceived and designed the overall research framework. P.T. developed the computational models and implemented the algorithms. J.J. performed the experiments and pre-processed the datasets. X.C. contributed to data analysis, result interpretation, and visualization. B.L. supervised the project, coordinated the research activities, and provided critical revisions. B.L. and P.T. drafted the manuscript. All authors reviewed and approved the final version of the manuscript.

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Competing interests

The authors declare no competing interests.

Data availability

The datasets used in this study are publicly available from the International Skin Imaging Collaboration (ISIC) archive (https://challenge.isic-archive.com/). All dermoscopic images and corresponding lesion annotations employed in our experiments are derived from the ISIC 2016, ISIC 2017, and ISIC 2018 datasets. No proprietary or patient-identifiable data were collected or used in this work. The experiments can be fully reproduced using these publicly accessible datasets.